Data-Backed Journal-Style Replication of *Effects of Search Timing on Innovation*

Replication Paper Project

2026-04-04

Abstract

This paper presents a local calculated replication of Effects of Search Timing on Innovation: The Value of Not Being in Sync with Rivals using only processed datasets and result files available in the workspace. The replication preserves the source paper's core empirical logic while making every local substitution explicit. The current local batch contains 4,839 observations across 167 entities, and the batch headline estimate is -0.0854 for ahead vs sync on event quality with p = 0.1178. The local event-level analogue produces weaker timing effects than the source paper's argument would suggest, especially for event quality. The contribution is a journal-style replication manuscript that separates direct source claims from executable local evidence.

Keywords: search timing; rivals; ahead versus behind; event-level innovation; local calculated replication

1 Introduction

This replication begins from a paper-first reading of Effects of Search Timing on Innovation: The Value of Not Being in Sync with Rivals and treats the source article as an empirical argument that has to be rebuilt before it can be judged. The motivating question is: The source paper asks whether moving ahead of, in sync with, or behind rivals changes innovative output and quality. The study is worth revisiting because The replication matters because timing relative to rivals is

a relational construct that is easy to cite and hard to rebuild transparently from local event data. The original article does not rest on a descriptive regularity alone. It ties a theory about search, differentiation, timing, or organizational learning to a specific unit of analysis, sample frame, and estimand. A careful replication therefore has to recover the structure of that logic before any local coefficients are interpreted. The executable evidence base used in this manuscript contains 4,839 observations across 167 entities and spans 1986 to 2016 in the local executable panel.. Those numbers are not filler. They define the actual inferential scope of the local reconstruction and set the boundary on what can be claimed honestly from the available datasets. The paper's contribution is an explicit event-level reconstruction of ahead, sync, and behind timing that shows what the timing mechanism looks like in a local patent-based analogue. The paper is written to read like a formal journal article rather than an internal workflow memo. The main text develops the theoretical logic, data reconstruction, derivations, empirical design, and results in continuous prose, while the appendices preserve the audit trail required for reruns and later extensions.

2 Original Study, Theory, and Replication Boundary

The focal source article is [1]. The original study can be summarized as a mechanism paper with a sharp empirical core. Timing relative to rivals is the substantive core of the original design. The original paper studies both innovation quantity and innovation quality. Ahead, sync, and behind are relational timing categories. That combination matters because the source article asks readers to believe not merely that two variables move together, but that a particular organizational or strategic mechanism explains why the relationship should appear in the data. In a setting like this, faithful replication requires more than reusing a title and rerunning a regression. The theoretical contribution lives in the way constructs are defined, how they enter the estimand, and which comparison the model is supposed to adjudicate. For that reason, this manuscript first reconstructs the published theory and only then turns to the local executable analogue. The replication boundary is therefore explicit. Direct source facts are carried over when they can be read from the published article itself. Local replication decisions appear only when the current workspace must translate the source design into verified processed datasets. The goal is disciplined

mechanism recovery, not rhetorical imitation that quietly changes the design.

2.1 Direct Source Facts Versus Replication Decisions

The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Timing relative to rivals is central. The local reconstruction therefore adopts the following bounded decision: The local event file reconstructs timing from recombination-combination first use. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline. The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Both quantity and quality are studied. The local reconstruction therefore adopts the following bounded decision: Event patent counts and mean inventive impact are the executable outcomes. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline. The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that The categories are ahead, sync, and behind. The local reconstruction therefore adopts the following bounded decision: Events without rival comparators are excluded because timing categories cannot be assigned. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline.

DirectSourceFact	LocalReplicationDecision
Timing relative to rivals is central.	The local event file reconstructs timing from recombination-combination first use.
Both quantity and quality are studied.	Event patent counts and mean inventive impact are the executable outcomes.
The categories are ahead, sync, and behind.	Events without rival comparators are excluded because timing categories cannot be assigned.

Direct source facts and explicit local execution decisions carried in the body of the paper.

3 Data, Sample, and Measures

The local evidence chain is deliberately modular. The source paper is treated as the authority for the research question, theoretical mechanism, and published design. Estimation, by contrast, proceeds only through the verified local files. The required dataset types in this paper are Event-level dataset built from first observed firm use of each recombination combination. Timing categories created by comparing event year against rival first-use years. Event outcomes measured by patent count and mean inventive impact. Operationally, the reconstruction moves through the following file chain: Analysis panel is backed by the local file katila_chen_2008_analysis_dataset.csv; Coefficient export is backed by the local file katila_chen_2008_model_coefficients.csv; Sample summary is backed by the local file katila_chen_2008_sample_summary.csv. The resulting executable panel supplies 4,839 observations across 167 entities and covers 1986 to 2016 in the local executable panel.. Every numeric claim in the results section is intended to trace back to one of those verified files rather than to a hand-copied number from the article. This data-lineage discipline matters for

two reasons. First, it prevents the manuscript from smuggling unverified preprocessing steps into the design. Second, it forces every substitution to be visible. If a construct cannot be reproduced from the current processed datasets, the paper states the analogue directly instead of pretending that the original archive has been fully recreated. The local analogue reconstructs first-use events for recombination combinations and classifies each event relative to rival timing. The local sample frame is First-use recombination events with rival-comparable timing categories in the local event panel. The observed time window in the executable panel is 1986 to 2016 in the local executable panel. The outcome is Event patent count and event mean inventive impact. The main executable panel used in the paper is katila_chen_2008_analysis_dataset.csv and the principal coefficient export is katila_chen_2008_model_coefficients.csv .

File	Role
katila_chen_2008_analysis_dataset.csv	Event-level timing panel
katila_chen_2008_model_coefficients.csv	Quantity and quality coefficient table
katila_chen_2008_sample_summary.csv	Timing-group raw means

Core local datasets and result files used in the current paper.

3.1 Executable File Selection and Data Availability

The present manuscript works from the processed datasets and calculated exports already available in the workspace for Katila and Chen (2008). That does not mean the paper skips data analysis. It means the data analysis begins from the current executable panel, current coefficient export, and current sample-summary files rather than from a freshly rebuilt raw archive. The substantive obligation is therefore to explain exactly why each verified file is needed and how it enters the paper's argument. The analysis panel used in this paper is katila_chen_2008_analysis_dataset.csv. Its role is to estimate the local model and regenerate descriptive statistics. This file is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. By naming the dataset rather than print-

ing a filesystem path, the paper keeps the emphasis on evidence and function rather than on local machine details. The coefficient export used in this paper is katila_chen_2008_model_coefficients.csv. Its role is to recover the reported local coefficients and p-values. This file is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. By naming the dataset rather than printing a filesystem path, the paper keeps the emphasis on evidence and function rather than on local machine details. The sample summary used in this paper is katila_chen_2008_sample_summary.csv. Its role is to audit sample construction and raw grouped moments. This file is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. By naming the dataset rather than printing a filesystem path, the paper keeps the emphasis on evidence and function rather than on local machine details. Taken together, these files form the minimum executable evidence chain behind the paper. The theory comes from the source article, the estimable sample comes from the processed analysis panel, the regression narrative comes from the current coefficient export, and the sample audit comes from the local summary file. That is the basis on which the manuscript makes every empirical claim.

Stage	NeededFor	Dataset	Exists	Status
Source article	Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices	katilachenasq2008.pdf	Yes	Direct source
Analysis panel	Estimate the local model and regenerate descriptive statistics	katila_chen_2008_analysis_dataset.csv	Yes	Executable local dataset
Coefficient export	Recover the reported local coefficients and p-values	katila_chen_2008_model_coefficients.csv	Yes	Executable local result file

Stage	NeededFor	Dataset	Exists	Status
Sample summary	Audit sample construction and raw grouped moments	katila_chen_2008_sample_summary.csv	Yes	Executable local result file

Verified current files used to support the paper's calculations.

3.2 Sample Reconstruction

The event-level panel partitions observations into ahead, sync, and behind categories. The largest category is behind with 3,034 events. Ahead events average 1.0705 patents and mean impact 4.5928, whereas behind events average 1.0906 patents and mean impact 4.4235. Sync events, the reference category, average 1.1261 patents and mean impact 4.9553. The descriptive timing contrast shows that the local event environment is not dominated by dramatic first-mover advantages before controls are added. That makes the later regression analysis substantively useful, because it asks whether timing still matters after accounting for event breadth and accumulated field experience.

Timing	Events	MeanEventPatents	MeanEventImpact
sync	444.0000	1.1261	4.9553
behind	3034.0000	1.0906	4.4235
ahead	1361.0000	1.0705	4.5928

Sample reconstruction from the current local dataset.

3.3 Variable Construction and Derivation

The construct Search timing is operationalized locally as Ahead / sync / behind relative to rivals and appears in the executable dataset as timing. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Compare each firm's first-use year for a combination to

rivals' first-use years.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity. The construct Innovation quantity is operationalized locally as Event patent count and appears in the executable dataset as event_patents. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Count patents inside each event.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity. The construct Innovation quality is operationalized locally as Event mean inventive impact and appears in the executable dataset as event_mean_impact. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Average patent-level impact inside each event.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity. The construct Search breadth is operationalized locally as Event breadth and appears in the executable dataset as event_breadth. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Carry forward the local event breadth measure.. This mapping is retained because it preserves

the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity. The construct Field experience is operationalized locally as Event field experience and appears in the executable dataset as event_field_exp. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Carry forward the local event field-experience measure.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity.

OriginalConstruct	LocalMeasure	ExactVariable	Derivation
Search timing	Ahead / sync / behind relative to rivals	timing	Compare each firm's first-use year for a combination to rivals' first-use years.
Innovation quantity	Event patent count	event_patents	Count patents inside each event.

OriginalConstruct	LocalMeasure	ExactVariable	Derivation
Innovation quality	Event mean inventive impact	event_mean_impact	Average patent-level impact inside each event.
Search breadth	Event breadth	event_breadth	Carry forward the local event breadth measure.
Field experience	Event field experience	event_field_exp	Carry forward the local event field-experience measure.

Variable mapping and derivation logic in the main text.

3.4 Construct-Level Derivation Logic

The derivation appendix records how each major construct enters the executable estimand for Katila and Chen (2008). The point is not to rehearse notation for its own sake, but to show how the source paper's conceptual vocabulary is translated into verified columns in the local panel. For Search timing, the manuscript uses Ahead / sync / behind relative to rivals under the exact local implementation timing. The derivation step is: Compare each firm's first-use year for a combination to rivals' first-use years.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Innovation quantity, the manuscript uses Event patent count under the exact local implementation event_patents. The derivation step is: Count patents inside each event.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Innovation quality, the manuscript uses Event mean inventive impact under the exact local implementation event_mean_impact. The derivation step is: Average patent-

level impact inside each event.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Search breadth, the manuscript uses Event breadth under the exact local implementation event_breadth. The derivation step is: Carry forward the local event breadth measure.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Field experience, the manuscript uses Event field experience under the exact local implementation event_field_exp. The derivation step is: Carry forward the local event field-experience measure.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept.

4 Descriptive Analysis of the Current Dataset

The current dataset yields a richer descriptive picture than a single mean or coefficient can convey. Event patents has mean 1.0882, standard deviation 0.4049, and an interquartile range from 1.0000 to 1.0000; Event total impact has mean 5.0087, standard deviation 8.2191, and an interquartile range from 1.0000 to 6.0000; Event mean impact has mean 4.5199, standard deviation 6.8945, and an interquartile range from 1.0000 to 6.0000; Event breadth has mean 2.4650, standard deviation 0.7319, and an interquartile range from 2.0000 to 3.0000 . The time-series view is equally informative. V1 changes from 274.0000 in the first observed period to 46.0000 in the last observed period; V2 changes from 1.1168 in the first observed period to 1.2174 in the last observed period; V3 changes from 4.1745 in the first observed period to 1.5362 in the last observed period . These temporal shifts help explain why a paper-comparable replication has to describe the support of the data before turning to regression output. The grouped event analysis also carries substantive information before estimation. Ahead events average 1.0705 patents and mean impact 4.5928, sync events average 1.1261 patents and mean impact 4.9553, and behind events average 1.0906 patents and mean impact 4.4235. The main models therefore build on a visible cross-group pattern rather than on a black-box regression alone. The correlation structure is reported in a compact form because it helps distinguish whether the main explanatory variables move in lockstep with the controls or whether the regression is drawing on independent variation. That is especially important

in these replications, where construct translation can otherwise make the specification look more deterministic than it really is.

4.1 Data Quality, Coverage, and Support

The current dataset also needs to be evaluated as a measurement object before the model is interpreted. Event patents has missing share 0.00; Event total impact has missing share 0.00; Event mean impact has missing share 0.00 . At the other end of the support, Event patents remains close to complete with missing share 0.00; Event total impact remains close to complete with missing share 0.00; Event mean impact remains close to complete with missing share 0.00 . This matters because the credibility of the regression depends on whether the focal constructs survive sample cleaning with enough support to be interpreted substantively. Coverage across entities is also visible in the current panel rather than assumed. 6066 contributes 393 observations; 5047 contributes 275 observations; 7585 contributes 202 observations; 10499 contributes 200 observations; 4087 contributes 182 observations . The purpose of reporting these concentrations is to show whether the estimated mechanism is being learned from a broad panel or from a narrow set of heavily observed entities. The support of the focal regressor is broad enough to permit a distributional reading in addition to the formal model. In the lowest observed quantile, the mean regressor is 2.2909 and the mean outcome is 4.4410 across 4,411 observations. In the highest observed quantile, the corresponding values are 4.2593, 5.3328, and 428 observations. This shift matters because it shows that the regression is being estimated on economically different parts of the observed support rather than on nearly identical cells. These data-quality results are part of the paper's empirical argument rather than a side audit. The replication matters because timing relative to rivals is a relational construct that is easy to cite and hard to rebuild transparently from local event data. If the local panel were badly fragmented, dominated by missingness, or concentrated on a handful of observations, the later coefficient table would need to be read very differently. Bringing this evidence into the body of the article is therefore necessary for an honest replication.

Variable	Missing	MissingShare
Event patents	0.0000	0.0000

Variable	Missing	MissingShare
Event total impact	0.0000	0.0000
Event mean impact	0.0000	0.0000
Event breadth	0.0000	0.0000
Event field experience	0.0000	0.0000

Missingness profile for variables analyzed in the body of the paper.

Variable	N	Mean	SD	P25	Median	P75	Min	M
Event patents	4839.0000	1.0882	0.4049	1.0000	1.0000	1.0000	1.0000	13.00
Event total impact	4839.0000	5.0087	8.2191	1.0000	3.0000	6.0000	0.0000	200.00
Event mean impact	4839.0000	4.5199	6.8945	1.0000	3.0000	6.0000	0.0000	200.00
Event breadth	4839.0000	2.4650	0.7319	2.0000	2.0000	3.0000	2.0000	10.00
Event field experience	4839.0000	2.7288	1.4265	1.7918	2.7081	3.6889	0.0000	7.86

Main descriptive statistics discussed in the body of the paper.

Period	V1	V2	V3	V4	V5
1986.0000	274.0000	1.1168	4.1745	2.3431	1.7241
1987.0000	313.0000	1.0767	3.8496	2.3642	2.0088
1988.0000	354.0000	1.1073	4.2290	2.2938	2.2682
1989.0000	411.0000	1.1168	3.8925	2.3382	2.3748
1990.0000	351.0000	1.1111	4.4128	2.3875	2.4820
1991.0000	351.0000	1.0855	4.6899	2.4444	2.6314
1992.0000	318.0000	1.0755	5.4591	2.4686	2.5444
1993.0000	321.0000	1.0903	5.5696	2.4984	2.7435

Period	V1	V2	V3	V4	V5
1994.0000	304.0000	1.0526	5.4940	2.5164	2.7160
1995.0000	242.0000	1.0826	6.3399	2.4917	2.8292
1996.0000	126.0000	1.0397	5.7262	2.5635	2.8326
1997.0000	111.0000	1.0811	5.5676	2.4685	3.0879
1998.0000	58.0000	1.0862	5.5891	2.4310	2.7553
1999.0000	138.0000	1.0725	5.7114	2.5652	2.8623
2000.0000	119.0000	1.0840	6.4762	2.4958	2.8887
2001.0000	94.0000	1.0319	4.0426	2.3936	3.0980
2002.0000	92.0000	1.0652	4.1467	2.7065	2.9604
2003.0000	82.0000	1.0122	4.4024	2.4756	3.3064
2004.0000	37.0000	1.0000	3.0541	2.4595	3.1022
2005.0000	43.0000	1.3256	5.6592	2.5116	3.0457
2006.0000	36.0000	1.0556	3.0694	2.5278	3.5952
2007.0000	38.0000	1.0000	2.7368	2.5000	3.3992
2008.0000	79.0000	1.2025	2.2443	2.6076	3.4918
2009.0000	24.0000	1.0417	2.5000	2.9167	3.5316
2010.0000	54.0000	1.0370	1.8889	2.9074	3.4028
2011.0000	75.0000	1.0400	4.3467	2.6133	4.0232
2012.0000	101.0000	1.0891	2.8383	2.6139	4.0222
2013.0000	78.0000	1.1026	2.4679	2.6795	3.5961
2014.0000	69.0000	1.1449	2.5352	2.8261	4.0339
2015.0000	100.0000	1.0300	1.9700	2.5400	4.0590
2016.0000	46.0000	1.2174	1.5362	2.6087	3.8081

Observed time patterns in the current dataset.

Timing	Events	MeanEventPatents	MeanEventImpact
sync	444.0000	1.1261	4.9553

Timing	Events	MeanEventPatents	MeanEventImpact
behind	3034.0000	1.0906	4.4235
ahead	1361.0000	1.0705	4.5928

Grouped summaries used to describe raw patterns in the current dataset.

Variable	Event patents	Event total impact	Event mean impact	Event breadth	Event field experience
Event patents	1.000	0.313	0.032	-0.025	0.089
Event total impact	0.313	1.000	0.915	0.065	0.002
Event mean impact	0.032	0.915	1.000	0.074	-0.027
Event breadth	-0.025	0.065	0.074	1.000	-0.242
Event field experience	0.089	0.002	-0.027	-0.242	1.000

Compact correlation structure for the main variables.

Entity	Observations	FirstPeriod	LastPeriod
6066.0000	393.0000	1986.0000	2016.0000
5047.0000	275.0000	1986.0000	2015.0000
7585.0000	202.0000	1986.0000	2012.0000
10499.0000	200.0000	1986.0000	2016.0000
4087.0000	182.0000	1986.0000	2013.0000
5073.0000	168.0000	1986.0000	2007.0000
4194.0000	151.0000	1986.0000	2010.0000
1300.0000	126.0000	1986.0000	2015.0000
1581.0000	117.0000	1986.0000	2004.0000
4060.0000	117.0000	1986.0000	2016.0000
4839.0000	113.0000	1986.0000	2016.0000
5606.0000	89.0000	1986.0000	2000.0000

Entity	Observations	FirstPeriod	LastPeriod
4503.0000	87.0000	1986.0000	2013.0000
10983.0000	82.0000	1986.0000	2015.0000
11436.0000	82.0000	1986.0000	1997.0000
7475.0000	81.0000	1986.0000	1995.0000
7435.0000	76.0000	1986.0000	2015.0000
12141.0000	72.0000	1995.0000	2016.0000
1690.0000	70.0000	1990.0000	2016.0000
2285.0000	64.0000	1986.0000	2015.0000

Most observed entities in the current dataset.

5 Empirical Strategy and Identification

The estimand targeted in this manuscript is Event-level models comparing ahead and behind timing categories with the sync benchmark while controlling for event breadth and field experience. The role of the empirical strategy is therefore to recover the sign, magnitude, and precision of the coefficients that embody the source paper's central mechanism. Identification remains observational. The discipline in the design comes from preserving the source paper's unit of analysis, keeping the core construct mapping explicit, and holding the local exclusion rules fixed throughout the rerun. This does not convert the study into an experimental design, but it does make the inferential claim narrower and more honest. The first set of replication decisions concerns scope. The unit of analysis is Search event, the sample frame is First-use recombination events with rival-comparable timing categories in the local event panel., and the empirical window is 1986 to 2016 in the local executable panel.. These choices define what counts as a comparable observation in the local reconstruction and prevent the paper from drifting into a broader but theoretically mismatched sample. The second set of decisions concerns measurement and specification. The outcome is Event patent count and event mean inventive impact., and the main explanatory design is Event-level models comparing ahead and behind timing categories to the sync benchmark.. This is where the replication

decides whether a variable is close enough to the published construct to remain in the estimand without changing the paper's substantive question. The third set of decisions concerns statistical discipline. Controls and fixed effects are handled as follows: Event breadth and field-experience controls, with available firm and year structure carried into estimation.. The crucial exclusion or substitution rule is: Events without a valid rival timing comparison are excluded because ahead, sync, and behind cannot be assigned honestly.. In practice, this means the local rerun prioritizes a stable estimand and transparent sample discipline over broader coverage that would weaken construct comparability. A final design choice concerns interpretation. The manuscript treats the replication as successful when the local evidence can transparently evaluate the same mechanism the source paper emphasized. When the local data architecture differs, the paper marks the result as a calculated analogue rather than claiming byte-identical duplication.

DecisionPoint	LocalSpecification
Research question	The source paper asks whether moving ahead of, in sync with, or behind rivals changes innovative output and quality.
Unit of analysis	Search event
Sample frame	First-use recombination events with rival-comparable timing categories in the local event panel.
Time window	1986 to 2016 in the local executable panel.
Outcome definition	Event patent count and event mean inventive impact.

DecisionPoint	LocalSpecification
Main explanatory design	Event-level models comparing ahead and behind timing categories to the sync benchmark.
Controls and fixed effects	Event breadth and field-experience controls, with available firm and year structure carried into estimation.
Exclusion or substitution logic	Events without a valid rival timing comparison are excluded because ahead, sync, and behind cannot be assigned honestly.

Replication decision register retained in the main text.

5.1 Why These Design Decisions Are Necessary

Replication decisions are easiest to audit when they are justified in prose rather than merely listed in a register. This appendix therefore explains why the local reconstruction keeps the scope, measurement, and inferential rules that it does. Scope decisions: the manuscript works at the level of Search event inside the sample frame First-use recombination events with rival-comparable timing categories in the local event panel. over 1986 to 2016 in the local executable panel.. These are not cosmetic constraints. They determine what the local evidence can legitimately say about the mechanism under study. Measurement decisions: the outcome is Event patent count and event mean inventive impact. and the main explanatory design is Event-level models comparing ahead and behind timing categories to the sync benchmark.. The local paper keeps these objects because they

preserve the estimand's substantive content while remaining executable from processed datasets already present in the workspace. Inference decisions: Event breadth and field-experience controls, with available firm and year structure carried into estimation. The main exclusion or substitution rule is Events without a valid rival timing comparison are excluded because ahead, sync, and behind cannot be assigned honestly. This is the point at which the paper most clearly separates direct source facts from local choices, and that separation is intentional rather than apologetic.

$$Y_e = \beta_1 \mathbb{1}[\text{Ahead}_e] + \beta_2 \mathbb{1}[\text{Behind}_e] + \beta_3 \text{EventBreadth}_e + \beta_4 \text{EventFieldExp}_e + \alpha_f + \gamma_t + \varepsilon_e. \quad (1)$$

6 Results

6.1 Main Estimates

The central local estimate is -0.0854 on ahead vs sync on event quality (p = 0.1178). The local event-level analogue produces weaker timing effects than the source paper's argument would suggest, especially for event quality. Event breadth is estimated at 0.1289 with p = ¡0.001. Ahead versus sync is estimated at -0.0854 with p = 0.1178. Behind versus sync is estimated at -0.0704 with p = 0.1208. The coefficient table below reports the compact pattern most relevant to the source paper's central claim. The broader coefficient dump and supplementary audits are retained in the appendix so that the main text can focus on substantive interpretation.

model	Term	estimate	std_error	p_value
katila_chen_event_quality	Ahead versus sync	-0.0854	0.0530	0.1178
katila_chen_event_quality	Behind versus sync	-0.0704	0.0441	0.1208
katila_chen_event_quality	Event breadth	0.1289	0.0195	0.0000
katila_chen_event_quality	Event field experience	0.0147	0.0132	0.2721

Key coefficient table used in the main text.

6.2 Detailed Interpretation of the Local Results

The coefficient pattern is not being summarized as a bare sign check. Event breadth is estimated at 0.1289 with a p-value of ¡0.001; Ahead versus sync is estimated at -0.0854 with a p-value of 0.1178; Behind versus sync is estimated at -0.0704 with a p-value of 0.1208; Event field experience is estimated at 0.0147 with a p-value of 0.2721 . Within the focal specification, the sharpest local signal comes from Event breadth, which is estimated at 0.1289 with p = ¡0.001. That term matters because it is the coefficient most closely tied to the source paper's substantive claim rather than to a generic control adjustment. The scale of the underlying variables also matters for interpretation. The outcome Event mean impact has mean 4.5199 and standard deviation 6.8945 in the local panel, while the main regressor Event breadth has mean 2.4650 and standard deviation 0.7319. A simple grouped comparison also helps clarify the pattern. Across the regressor quantiles, the mean outcome ranges from 4.4410 to 5.3328 while the mean regressor ranges from 2.2909 to 4.2593. This does not substitute for the model, but it shows how much raw variation the specification must explain. Taken together, the descriptive evidence and formal estimates should be read as a structured assessment of the paper's mechanism. The local event-level analogue produces weaker timing effects than the source paper's argument would suggest, especially for event quality. The substantive question is not whether every coefficient matches a published table entry digit for digit, but whether the local evidence supports the same directional and conceptual interpretation.

6.3 Coefficient Families in the Current Estimation Output

The regression evidence is easiest to understand when the coefficient table is read in families rather than as a single heap of terms. The paper therefore separates the estimates by model and highlights the strongest positive, strongest negative, and most focal signals within each specification. Within katila_chen_event_patents, the leading coefficient family reads as follows: Event field experience = 0.0360 (p = ¡0.001); Behind versus sync = -0.0490 (p = 0.0114); Ahead versus sync = -0.0291 (p = 0.1416); Event breadth = 0.0079 (p = 0.2042). The largest positive estimate in this specification is Event field experience at 0.0360, while the largest negative estimate is Behind versus sync at -0.0490. Reading the specification this way matters because it shows which part of the model is

doing the explanatory work and which part is mainly serving as adjustment structure. Within katila_chen_event_quality, the leading coefficient family reads as follows: Event breadth = 0.1289 (p = ¡0.001); Ahead versus sync = -0.0854 (p = 0.1178); Behind versus sync = -0.0704 (p = 0.1208); Event field experience = 0.0147 (p = 0.2721). The largest positive estimate in this specification is Event breadth at 0.1289, while the largest negative estimate is Ahead versus sync at -0.0854. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure. This fuller coefficient reading is especially important for Katila and Chen (2008) because the source paper's contribution is not exhausted by one headline estimate. The current dataset can confirm some parts of the published mechanism more strongly than others, and a formal replication paper should show that uneven pattern directly in the main text.

6.4 Model Comparison Using the Current Dataset

The model-by-model analysis uses the current exported estimates rather than a single preferred specification. In katila_chen_event_patents, the most informative focal term is Event field experience with estimate 0.0360 and p = ¡0.001; In katila_chen_event_quality, the most informative focal term is Event breadth with estimate 0.1289 and p = ¡0.001 . The katila_chen_event_patents specification contributes its own reading to the replication. Event field experience is estimated at 0.0360 with p = ¡0.001 and is classified as a strong signal; Behind versus sync is estimated at -0.0490 with p = 0.0114 and is classified as a moderate signal; Ahead versus sync is estimated at -0.0291 with p = 0.1416 and is classified as a weak signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient. The katila_chen_event_quality specification contributes its own reading to the replication. Event breadth is estimated at 0.1289 with p = ¡0.001 and is classified as a strong signal; Ahead versus sync is estimated at -0.0854 with p = 0.1178 and is classified as a weak signal; Behind versus sync is estimated at -0.0704 with p = 0.1208 and is classified as a weak signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient. This model

comparison matters for Katila and Chen (2008) because the source article's claim can only be evaluated responsibly if the pattern persists across the outcome or specification variants that are actually present in the local results folder. Put differently, the current dataset should be allowed to show where the mechanism is stable, where it weakens, and where it changes sign under a different outcome definition.

Term	Model	Estimate	PValue	Signal	Reading
Ahead versus sync	katila_chen_event_patents	-0.0291	0.1416	Weak	Negative estimate with p = 0.1416.
Behind versus sync	katila_chen_event_patents	-0.0490	0.0114	Moderate	Negative estimate with p = 0.0114.
Event breadth	katila_chen_event_patents	0.0079	0.2042	Weak	Positive estimate with p = 0.2042.
Event field experience	katila_chen_event_patents	0.0360	0.0000	Strong	Positive estimate with p = ¡0.001.
Ahead versus sync	katila_chen_event_quality	-0.0854	0.1178	Weak	Negative estimate with p = 0.1178.

Term	Model	Estimate	PValue	Signal	Reading
Behind versus sync	katila_chen_event_quality	-0.0704	0.1208	Weak	Negative estimate with p = 0.1208.
Event breadth	katila_chen_event_quality	0.1289	0.0000	Strong	Positive estimate with p = ¡0.001.
Event field experience	katila_chen_event_quality	0.0147	0.2721	Weak	Positive estimate with p = 0.2721.

Model-by-model interpretation table drawn from the current coefficient exports.

6.5 Diagnostic Evidence

A separate model-fit export is not available for this paper, so the interpretation relies on the coefficient file and the observed distribution of the current dataset.

term	estimate	std_error	statistic	p_value	model
timing::ahead	-0.0291	0.0198	-1.4698	0.1416	katila_chen_event_pate
timing::behind	-0.0490	0.0194	-2.5304	0.0114	katila_chen_event_pate
event_breadth	0.0079	0.0062	1.2696	0.2042	katila_chen_event_pate
event_field_exp	0.0360	0.0063	5.7486	0.0000	katila_chen_event_pate
timing::ahead	-0.0854	0.0530	-1.6102	0.1178	katila_chen_event_qual
timing::behind	-0.0704	0.0441	-1.5969	0.1208	katila_chen_event_qual
event_breadth	0.1289	0.0195	6.6249	0.0000	katila_chen_event_qual

term	estimate	std_error	statistic	p_value	model
event_field_exp	0.0147	0.0132	1.1188	0.2721	katila_chen_event_qual

Full coefficient table from the current local estimation output.

7 Comparison to the Published Study

The comparison to the published paper is intentionally narrow. The current local event panel does not yield strong evidence that being ahead of rivals improves event quality. The ahead coefficient is negative and weak in the quality model, while being behind is more clearly negative for event patent counts. The manuscript therefore evaluates concordance at the level of mechanism, sign pattern, and construct behavior before drawing any conclusion about exact numerical replication. This comparison rule matters because the local paper is based only on processed datasets and verified outputs already present in the workspace. When the original paper depends on proprietary, historical, or otherwise unavailable preprocessing layers, a full byte-identical recreation is not claimed. The more defensible question is whether the local evidence would lead a careful reader to a similar substantive interpretation.

8 Discussion

The current local event panel does not yield strong evidence that being ahead of rivals improves event quality. The ahead coefficient is negative and weak in the quality model, while being behind is more clearly negative for event patent counts. Taken together, the results are most informative about whether the source paper's mechanism is directionally recoverable in local data rather than whether every original coefficient can be duplicated exactly. The first limitation concerns measurement. The local event environment is a patent-recombination analogue rather than the source paper's original innovation-search setting. The second limitation concerns sample translation. Some combinations have sparse rival overlap, which limits timing variation. The third limitation concerns

interpretation. The estimates are best read as a calculated analogue of the timing mechanism.

9 Conclusion

The paper's contribution is an explicit event-level reconstruction of ahead, sync, and behind timing that shows what the timing mechanism looks like in a local patent-based analogue. The paper therefore contributes a traceable local replication record that can be rerun, audited, and extended as additional datasets or original-source details become available.

A Appendix A: Supplementary Sample Reconstruction

The sample table below shows the observable route from the broader local file universe to the final estimating sample. It is included to make sample narrowing visible and to show where exclusions enter the paper.

Timing	Events	MeanEventPatents	MeanEventImpact
sync	444.0000	1.1261	4.9553
behind	3034.0000	1.0906	4.4235
ahead	1361.0000	1.0705	4.5928

Detailed sample reconstruction table.

B Appendix B: Supplementary Descriptive Tables

The descriptive appendix documents the scale, spread, and support of the variables used in the paper. These moments matter because they determine how the estimated coefficients should be read in substantive rather than merely statistical terms.

Variable	N	Mean	SD	P25	Median	P75	Min	M
Event patents	4839.0000	1.0882	0.4049	1.0000	1.0000	1.0000	1.0000	13.00
Event total impact	4839.0000	5.0087	8.2191	1.0000	3.0000	6.0000	0.0000	200.00
Event mean impact	4839.0000	4.5199	6.8945	1.0000	3.0000	6.0000	0.0000	200.00
Event breadth	4839.0000	2.4650	0.7319	2.0000	2.0000	3.0000	2.0000	10.00
Event field experience	4839.0000	2.7288	1.4265	1.7918	2.7081	3.6889	0.0000	7.86

Descriptive statistics appendix.

C Appendix C: Correlation Matrix

Variable	Event patents	Event total impact	Event mean impact	Event breadth	Event field experience
Event patents	1.000	0.313	0.032	-0.025	0.089
Event total impact	0.313	1.000	0.915	0.065	0.002
Event mean impact	0.032	0.915	1.000	0.074	-0.027
Event breadth	-0.025	0.065	0.074	1.000	-0.242
Event field experience	0.089	0.002	-0.027	-0.242	1.000

Correlation matrix for the main numeric variables.

D Appendix D: Yearly Audit

The yearly audit provides a simple check on whether the local panel is driven by a narrow time slice or whether the variables and sample counts evolve over a broad observed window.

Period	V1	V2	V3	V4	V5
1986.0000	274.0000	1.1168	4.1745	2.3431	1.7241
1987.0000	313.0000	1.0767	3.8496	2.3642	2.0088
1988.0000	354.0000	1.1073	4.2290	2.2938	2.2682
1989.0000	411.0000	1.1168	3.8925	2.3382	2.3748
1990.0000	351.0000	1.1111	4.4128	2.3875	2.4820
1991.0000	351.0000	1.0855	4.6899	2.4444	2.6314
1992.0000	318.0000	1.0755	5.4591	2.4686	2.5444
1993.0000	321.0000	1.0903	5.5696	2.4984	2.7435
1994.0000	304.0000	1.0526	5.4940	2.5164	2.7160
1995.0000	242.0000	1.0826	6.3399	2.4917	2.8292
1996.0000	126.0000	1.0397	5.7262	2.5635	2.8326
1997.0000	111.0000	1.0811	5.5676	2.4685	3.0879
1998.0000	58.0000	1.0862	5.5891	2.4310	2.7553
1999.0000	138.0000	1.0725	5.7114	2.5652	2.8623
2000.0000	119.0000	1.0840	6.4762	2.4958	2.8887
2001.0000	94.0000	1.0319	4.0426	2.3936	3.0980
2002.0000	92.0000	1.0652	4.1467	2.7065	2.9604
2003.0000	82.0000	1.0122	4.4024	2.4756	3.3064
2004.0000	37.0000	1.0000	3.0541	2.4595	3.1022
2005.0000	43.0000	1.3256	5.6592	2.5116	3.0457
2006.0000	36.0000	1.0556	3.0694	2.5278	3.5952
2007.0000	38.0000	1.0000	2.7368	2.5000	3.3992
2008.0000	79.0000	1.2025	2.2443	2.6076	3.4918
2009.0000	24.0000	1.0417	2.5000	2.9167	3.5316

Period	V1	V2	V3	V4	V5
2010.0000	54.0000	1.0370	1.8889	2.9074	3.4028
2011.0000	75.0000	1.0400	4.3467	2.6133	4.0232
2012.0000	101.0000	1.0891	2.8383	2.6139	4.0222
2013.0000	78.0000	1.1026	2.4679	2.6795	3.5961
2014.0000	69.0000	1.1449	2.5352	2.8261	4.0339
2015.0000	100.0000	1.0300	1.9700	2.5400	4.0590
2016.0000	46.0000	1.2174	1.5362	2.6087	3.8081

Year-by-year sample audit.

E Appendix E: Grouped Outcome Patterns

Grouped summaries do not replace the formal model, but they show the raw outcome structure that the specification is attempting to explain. For that reason they are retained here as descriptive support rather than promoted to the status of a main estimate.

Quantile	N	MeanRegressor	MeanOutcome
1.0000	4411.0000	2.2909	4.4410
2.0000	428.0000	4.2593	5.3328

Quantile summary for the main regressor.

Timing	Events	MeanEventPatents	MeanEventImpact
sync	444.0000	1.1261	4.9553
behind	3034.0000	1.0906	4.4235
ahead	1361.0000	1.0705	4.5928

Raw group means from the local summary file.

F Appendix F: Full Coefficient Table and Fit Statistics

The full coefficient export is included for completeness so that readers can see the focal terms in the context of the broader specification. Where a fit file is available, it is appended here rather than in the main text to keep the article narrative readable.

term	estimate	std_error	statistic	p_value	model
timing::ahead	-0.0291	0.0198	-1.4698	0.1416	katila_chen_event_pate
timing::behind	-0.0490	0.0194	-2.5304	0.0114	katila_chen_event_pate
event_breadth	0.0079	0.0062	1.2696	0.2042	katila_chen_event_pate
event_field_exp	0.0360	0.0063	5.7486	0.0000	katila_chen_event_pate
timing::ahead	-0.0854	0.0530	-1.6102	0.1178	katila_chen_event_qual
timing::behind	-0.0704	0.0441	-1.5969	0.1208	katila_chen_event_qual
event_breadth	0.1289	0.0195	6.6249	0.0000	katila_chen_event_qual
event_field_exp	0.0147	0.0132	1.1188	0.2721	katila_chen_event_qual

Full coefficient dump exported from the local estimation script.

G Appendix G: Coverage Tables

Variable	Missing	MissingShare
Event patents	0.0000	0.0000
Event total impact	0.0000	0.0000
Event mean impact	0.0000	0.0000
Event breadth	0.0000	0.0000
Event field experience	0.0000	0.0000

Missingness audit for the variables summarized in the manuscript.

Entity	Observations	FirstPeriod	LastPeriod
6066.0000	393.0000	1986.0000	2016.0000
5047.0000	275.0000	1986.0000	2015.0000
7585.0000	202.0000	1986.0000	2012.0000
10499.0000	200.0000	1986.0000	2016.0000
4087.0000	182.0000	1986.0000	2013.0000
5073.0000	168.0000	1986.0000	2007.0000
4194.0000	151.0000	1986.0000	2010.0000
1300.0000	126.0000	1986.0000	2015.0000
1581.0000	117.0000	1986.0000	2004.0000
4060.0000	117.0000	1986.0000	2016.0000
4839.0000	113.0000	1986.0000	2016.0000
5606.0000	89.0000	1986.0000	2000.0000
4503.0000	87.0000	1986.0000	2013.0000
10983.0000	82.0000	1986.0000	2015.0000
11436.0000	82.0000	1986.0000	1997.0000
7475.0000	81.0000	1986.0000	1995.0000
7435.0000	76.0000	1986.0000	2015.0000

Entity	Observations	FirstPeriod	LastPeriod
12141.0000	72.0000	1995.0000	2016.0000
1690.0000	70.0000	1990.0000	2016.0000
2285.0000	64.0000	1986.0000	2015.0000

Top entities by local observation count.

References

[1] Riitta Katila and Eric L. Chen. Effects of search timing on innovation: The value of not being in sync with rivals. *Administrative Science Quarterly*, 2008.